A. Inference and Proof Structure

Argument, validity, and countermodel

An **argument** consists of premises and a conclusion. It is **valid** when every valuation or interpretation that makes all premises true also makes the conclusion true. A **countermodel** makes every premise true and the conclusion false; one countermodel proves invalidity.

Propositional rules of inference

$p,\ p \to q \therefore q$	Modus Ponens (MP)
$\neg q,\ p \to q \therefore \neg p$	Modus Tollens (MT)
$p \to q,\ q \to r \therefore p \to r$	Hypothetical Syllogism
$p \lor q,\ \neg p \therefore q$	Disjunctive Syllogism
$p,\ q \therefore p \land q$	Conjunction
$p \land q \therefore p$	Simplification
$p \therefore p \lor q$	Addition
$p \lor q,\ p \to r,\ q \to r \therefore r$	Proof by cases

A.4: propositional derivation

Problem. From $p \to q$, $q \to r$, and $\neg r$, derive $\neg p$. Name the rule used at each step.

Worked solution. Target: derive $\neg p$ from $p \to q$, $q \to r$, and $\neg r$. Start at $\neg r$ and move backward through the conditional chain:

1.	$p \to q$	premise
2.	$q \to r$	premise
3.	$\neg r$	premise
4.	$\neg q$	MT, 2 and 3
5.	$\neg p$	MT, 1 and 4.

Line 4 removes the possibility of q; line 5 then removes the possibility of p. Name both the rule and the lines used.

Conclusion. $\neg p$, obtained by two applications of modus tollens.

Quantifier rules

Universal instantiation (UI): from $\forall x\, P(x)$ infer $P(c)$ for any named object c in the domain.

Universal generalization (UG): after proving $P(c)$ for a completely arbitrary c, infer $\forall x\, P(x)$.

Existential instantiation (EI): from $\exists x\, P(x)$ introduce a fresh name a with $P(a)$. It denotes some witness, not an arbitrary object.

Existential generalization (EG): from $P(c)$ infer $\exists x\, P(x)$.

Universal and existential witnesses differ

Problem (A.5). Over a nonempty domain, derive $\forall x Q(x)$ from $\forall x(P(x) \to Q(x))$ and $\forall x P(x)$, naming each rule.

Worked solution: (1) let c be arbitrary; (2) UI gives $P(c) \to Q(c)$ and $P(c)$; (3) MP gives $Q(c)$; (4) because c was arbitrary, UG gives $\forall x Q(x)$. **Conclusion:** the desired universal statement follows.

Existential warning: from $\exists x P(x)$ and $\exists x Q(x)$, EI may introduce different witnesses a and b. You may not conclude $P(a) \land Q(a)$ or $\exists x(P(x) \land Q(x))$ without evidence that the witnesses coincide.

A.3: countermodel for reused witnesses

Problem. Over one common domain, identify the error in the proposed inference below. Then give a two-element domain and interpretations of P, Q making both premises true and the conclusion false: $\exists x P(x)$, $\exists x Q(x)$, therefore $\exists x(P(x) \land Q(x))$.

Worked solution. The error is reusing an existential witness without justification; the two premises may have different witnesses. **Build:** use domain $\{1, 2\}$; let P hold only of 1 and Q only of 2. **Check premises:** 1 witnesses $\exists x P(x)$, and 2 witnesses $\exists x Q(x)$. **Check conclusion:** neither object satisfies both predicates, so $\exists x(P(x) \land Q(x))$ is false. True premises and a false conclusion prove invalidity.

Conclusion. The witnesses need not be the same: P holds only at 1, and Q only at 2.

Invalid forms

$p \to q, q \therefore p$ affirms the consequent. $p \to q, \neg p \therefore \neg q$ denies the antecedent. Neither is valid. A conclusion may be true even when the displayed argument for it is invalid.

A readiness

Can you execute a short derivation with named rules, recognize invalid forms, use UI/EI with correct scopes, and construct a small countermodel when an existential witness has been reused?

B. Direct Proofs from Definitions

Definitions used on this Core

For integers n:

$$\begin{aligned} n \text{ is even} &\iff n = 2k \text{ for some } k \in \mathbb{Z}, \\ n \text{ is odd} &\iff n = 2k+1 \text{ for some } k \in \mathbb{Z}, \\ d \mid n &\iff n = dk \text{ for some } k \in \mathbb{Z}. \end{aligned}$$

A real r is **rational** iff $r = a/b$ for integers a, b with $b \neq 0$; otherwise it is **irrational**. $A \subseteq B$ means $\forall x(x \in A \to x \in B)$.

Direct-proof template

For "if P, then Q":

1. Let the objects be arbitrary subject to the stated domain.
2. Assume P.
3. Unpack P using definitions and introduce correctly scoped witnesses.
4. Calculate or reason to the defining form of Q.
5. State why the new witness belongs to the required domain.

B.1: an even-square proof

Problem. Give a direct proof that if an integer n is even, then n^2 is even.
Worked solution. Set up: let arbitrary $n \in \mathbb{Z}$ be even, so $n = 2k$ for some $k \in \mathbb{Z}$. **Calculate and expose a factor of 2:**

$$n^2 = (2k)^2 = 4k^2 = 2(2k^2).$$

Close: $2k^2 \in \mathbb{Z}$, so the display has form 2(integer). Thus n^2 is even. The new witness is $2k^2$.
Conclusion. $n^2 = 2(2k^2)$ is even because $2k^2 \in \mathbb{Z}$.

B.3: two odd integers

Problem. Give a direct proof that the sum of two odd integers is even.
Worked solution. Set up: let arbitrary odd integers a, b satisfy $a = 2j + 1$ and $b = 2k + 1$ for some $j, k \in \mathbb{Z}$. Use different witnesses. **Combine:**

$$a + b = 2j + 1 + 2k + 1 = 2(j + k + 1).$$

Close: $j + k + 1 \in \mathbb{Z}$, so $a + b$ is twice an integer and is even.
Conclusion. $a + b = 2(j + k + 1)$, with integer witness $j + k + 1$.

B.2: sum of rational numbers

Problem. Give a direct proof that the sum of two rational numbers is rational.
Worked solution. Set up: let $r, s \in \mathbb{Q}$. Choose integers a, b, c, d with $b, d \neq 0$, $r = a/b$, and $s = c/d$. **Use a common denominator:**

$$r + s = \frac{ad + bc}{bd}.$$

Close: $ad + bc, bd \in \mathbb{Z}$, and $bd \neq 0$. The displayed fraction satisfies the definition of rational; lowest terms are not required.
Conclusion. $r + s = (ad + bc)/(bd) \in \mathbb{Q}$, since numerator and denominator are integers and $bd \neq 0$.

B.5: direct element proof

Problem. Let A, B, C be sets. Give a direct element proof that if $A \subseteq B$, then $A \cap C \subseteq B \cap C$.
Worked solution. Assume $A \subseteq B$. **Start on the left:** let arbitrary $x \in A \cap C$. **Unpack:** $x \in A$ and $x \in C$. **Use the hypothesis:** $x \in A$ and $A \subseteq B$ give $x \in B$. **Repack:** $x \in B$ and $x \in C$ give $x \in B \cap C$. Since x was arbitrary, the subset claim follows.
Conclusion. Every $x \in A \cap C$ belongs to $B \cap C$, so $A \cap C \subseteq B \cap C$.

Necessary proof language

Arbitrary means no special property beyond the hypotheses was chosen. A **witness** is a specific object whose existence establishes an existential claim. A **lemma** is a proved supporting result. A **counterexample** satisfies a universal claim's hypotheses but violates its conclusion.

Common errors

Do not write $a = 2k$ and $b = 2k$ merely because both are even; their witnesses may differ. Do not divide by a variable without knowing it is nonzero. Do not give an example when the claim is universal. End by connecting the algebra to the exact definition requested.

B readiness

Can you identify the target definition, introduce independent witnesses, perform the central algebraic or element-chasing step, verify the new witness, and state the conclusion for arbitrary inputs?

C–D. Indirect Methods, Cases, Existence, and Refutation

Contrapositive and contradiction

The contrapositive of $P \to Q$ is $\neg Q \to \neg P$ and is logically equivalent to the original. A **proof by contradiction** assumes the negation of the desired conclusion and derives an impossibility such as $R \wedge \neg R$. State which method you are using.

C: contraposition workflow

Write the contrapositive explicitly, assume its antecedent, and prove its consequent. If the antecedent is a disjunction, use cases. Do not assume the original conclusion.

C.1: product odd implies both factors odd

Problem. Let $m, n \in \mathbb{Z}$. Prove by contraposition that if mn is odd, then both m and n are odd. State the contrapositive explicitly and handle its disjunction by cases.
Worked solution. Step 1: write the contrapositive: if m is even or n is even, then mn is even. **Step 2:** use cases. If $m = 2k$, then $mn = 2(kn)$; if $n = 2\ell$, then $mn = 2(m\ell)$. **Step 3:** in either case the quantity in parentheses is an integer, so mn is even. The contrapositive, hence the original claim, is true.
Conclusion. The contrapositive holds in both cases; therefore an odd product has two odd factors.

C.3: no largest integer

Problem. Prove by contradiction that there is no largest integer.
Worked solution. Assume the opposite: let N be a largest integer. **Construct:** $N+1$ is an integer and $N+1 > N$. **Contradiction:** maximality requires $N+1 \leq N$, while arithmetic gives $N+1 > N$. Therefore no largest integer exists.
Conclusion. No largest integer exists, because $N+1$ contradicts any proposed largest integer N.

C.4: rational plus irrational

Problem. Let r be rational and x irrational. Prove by contradiction that $r + x$ is irrational.
Worked solution. Let $r \in \mathbb{Q}$ and $x \notin \mathbb{Q}$. **Assume the opposite:** $r + x \in \mathbb{Q}$. **Isolate the irrational number:** rationals are closed under subtraction, so $x = (r+x) - r \in \mathbb{Q}$. This contradicts the hypothesis $x \notin \mathbb{Q}$; hence $r + x$ is irrational.
Conclusion. $r + x$ is irrational; otherwise subtracting r makes x rational.

Proof by cases and existence

A **proof by cases** partitions all possibilities and proves the claim in each exhaustive case. A **constructive existence proof** gives an explicit witness and verifies every required property. A **nonconstructive proof** establishes existence without identifying a witness.

D.1: cases on parity

Problem. Prove by cases that $n(n+1)$ is even for every integer n.
Worked solution. Exhaustive split: every integer n has form $2k$ or $2k+1$. **Even case:** $n(n+1) = 2[k(n+1)]$. **Odd case:**

$$n(n+1) = (2k+1)(2k+2) = 2[(2k+1)(k+1)].$$

Both bracketed quantities are integers. Since all integers occur in one case, $n(n+1)$ is always even.
Conclusion. In either parity case the product is twice an integer; hence it is even for every $n \in \mathbb{Z}$.

D.3: constructive existence

Problem. Give a constructive existence proof that there are four consecutive composite positive integers. Verify explicitly that every integer in your example is composite.
Worked solution. Witnesses: $24, 25, 26, 27$ are consecutive and positive. **Verification:**

$$24 = 2 \cdot 12, \quad 25 = 5 \cdot 5, \quad 26 = 2 \cdot 13, \quad 27 = 3 \cdot 9.$$

Each factorization is nontrivial, so all four are composite. Listing numbers without checking the required property would not finish the proof.
Conclusion. $24, 25, 26, 27$ are four consecutive composite positive integers.

D.4: counterexamples

Problem. Disprove each universal statement by a counterexample: (a) the sum of two irrational numbers is irrational; (b) $n^2 + n + 41$ is prime for every nonnegative integer n.
Worked solution. Irrational sum claim: choose $\sqrt{2}$ and $-\sqrt{2}$ (both irrational). Their sum is 0 (rational), so the hypothesis holds and the conclusion fails.
Prime-output claim: choose the allowed input $n = 40$: $40^2 + 40 + 41 = 41^2$, which is composite. One checked counterexample refutes a universal claim.
Conclusion. (a) $\sqrt{2} + (-\sqrt{2}) = 0$; (b) at $n = 40$, the value is 41^2.

Common errors

Cases must be exhaustive. A counterexample must obey every domain restriction. In contradiction, do not merely restate the assumed negation; derive a genuine conflict. In contraposition, negate both sides correctly and reverse the arrow.

C–D readiness

Can you state a contrapositive, run a contradiction, split an exhaustive case, verify a constructive witness, and refute a universal claim with a complete counterexample?

E. Strategy Choice and Proof Repair

Choose a promising method

Direct	Definitions turn the hypothesis into the target form.
Contrapositive	The negated conclusion has a useful structural definition.
Contradiction	The conclusion denies existence or involves irrationality.
Cases	The domain naturally partitions (parity, sign, membership).
Constructive	The claim asks only that an example exists.
Counterexample	The universal claim looks too strong.

State the first substantive step, not merely the method name.

E.2: repair reused witnesses

Problem. The following proof is intended to show that the sum of two even integers is even. Identify the gap and repair it.
"Let a and b be even. Then $a = 2k$ and $b = 2k$ for some integer k. Hence $a + b = 4k$, which is even."
Worked solution. Gap: "$a = 2k$ and $b = 2k$" forces $a = b$; a shared witness was not justified. **Repair:** choose $j, k \in \mathbb{Z}$ with $a = 2j$ and $b = 2k$. Then $a + b = 2(j + k)$, and $j + k \in \mathbb{Z}$, so the sum is even.
Conclusion. Use independent integer witnesses: $a = 2j$, $b = 2k$, and $a + b = 2(j + k)$.

E.3: repair division by zero

Problem. The following proof claims that the product of a rational number and an irrational number is irrational. Explain why it fails when $r = 0$. Add the weakest necessary restriction on r, state the corrected theorem, and prove it.
"Let r be rational and x irrational. If rx were rational, then $x = (rx)/r$ would be rational, a contradiction."
Worked solution. Gap: the original claim fails for $r = 0$, and dividing by r may be illegal: for instance, $r = 0$ and $x = \sqrt{2}$ give $rx = 0$, which is rational. **Correct statement:** if $r \in \mathbb{Q} \setminus \{0\}$ and x is irrational, then rx is irrational. **Proof:** if rx were rational, division by nonzero rational r would give $x = (rx)/r \in \mathbb{Q}$, a contradiction.
Conclusion. Require $r \neq 0$: a nonzero rational times an irrational is irrational.

Whole-Core readiness

Can you perform an inference, write a direct proof, use contraposition or contradiction, handle cases/existence/counterexamples, and select or repair a strategy? A live 20-minute form selects one complete problem from each section A–E; four fully correct problems earn Meets and five earn Exceeds.

E.4: repair circular reasoning

Problem. The following argument is circular. Explain the circularity, then replace it with a valid proof.
"To prove that n^2 is odd whenever the integer n is odd, assume n^2 is odd. Thus $n^2 = 2k + 1$ for some integer k, so n^2 is odd."
Worked solution. Gap: assuming n^2 is odd assumes the conclusion. **Repair:** start with the hypothesis $n = 2k + 1$. Then

$$n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1,$$

The parenthesized quantity is an integer, so the result has form $2(\text{integer}) + 1$ and n^2 is odd.
Conclusion. Begin with $n = 2k + 1$, not with the conclusion; $n^2 = 2(2k^2 + 2k) + 1$ is odd.

E.5: repair a bad starting element

Problem. The following proof is intended to show that $A \subseteq B$ whenever $A \cap B = A$. Identify the gap and repair it.
"Let $x \in A \cap B$. Then $x \in B$. Therefore $A \subseteq B$."
Worked solution. Gap: starting with $x \in A \cap B$ does not cover arbitrary $x \in A$. **Repair:** let arbitrary $x \in A$. The equality $A = A \cap B$ gives $x \in A \cap B$, so $x \in B$. Since x was arbitrary, $A \subseteq B$.
Conclusion. Start with arbitrary $x \in A$, then use $A = A \cap B$ to obtain $x \in B$.

Essential versus minor errors

An **essential error** breaks the proof: assuming the conclusion, reversing an implication, using an unjustified common witness, dividing by a possible zero, omitting a case, or giving an example instead of a universal proof. Minor notation is repairable when scope and reasoning remain unambiguous.

A proof-repair checklist

Ask:

1. Did the proof start with the correct arbitrary object or hypothesis?
2. Are every witness and variable scoped correctly?
3. Is any implication reversed or invalid inference used?
4. Is division known to be legal?
5. Are cases exhaustive and conclusions exact?
6. Does the final line invoke the requested definition?

Name the gap before replacing it.

Presentation matters

A proof is a readable argument, not a stack of equations. State assumptions, connect equations with words, identify the witness produced, and finish with the exact conclusion. A correct idea with a missing essential justification is not a complete proof.